

Energy device in the market, say, a smartphone, talk to a Bluetooth Mesh network without waiting for Google or Apple to upgrade their stack with new APIs. Standard APIs that are already available under those platforms can be used to talk to the rest of the network via a proxy node.

There are multiple proxy nodes all over the building, and standard Bluetooth Low Energy applications that work with the mesh network can be written. So, range of compatibility increases with every device. That is a major advantage of this technology.

## Bluetooth Low Energy versus Bluetooth Mesh

In the world of Bluetooth Low Energy, security is optional. It is all about two devices, and the question about the type of security required is something only the manufacturers can decide. It is like a toolbox; there are lots of security capabilities, but manufacturers decide

which of those capabilities they will use for their products.

Bluetooth Mesh is different because no manufacturer is allowed to reduce the security of the network. All messages are encrypted and authenticated. This means messages in the network cannot be spoofed, and so all communication is secure. It is possible to separate the security of different applications by using different application keys, one for each mesh application. Devices only get application keys that are relevant to the application they are a member of.

Networks can be divided into sub-nets and different areas can be isolated cryptographically. This is great for hotels where each individual room has its own sub-net key issued at the time a booking is made. Messages are also obfuscated. This is a privacy

mechanism, and network analysis attacks and figuring out patterns of communication are not possible.

The whole thing starts with the provisioning process, which is how devices get those keys in the first place.

Bluetooth Mesh is a major new wireless communication technology, which we are already seeing adopted in the emerging world of the smart building. **END** 

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There are many ways to add value to GIMP and to make your customers happy.

**Respect the GPL:** The GPL requires you to make the source code available. The best solution is to include the source code on the same medium as the GIMP installation package, but you can also include a written offer to supply the source code on request. Note that you cannot simply give a link to the GIMP FTP mirrors; it should be the exact source code that was used to compile the binary package that you are selling and you have to cover the costs of redistribution yourself. If you sell and distribute the binaries online, the GPL requires you to make the source code available “from the same place.” So giving a link to the GIMP mirrors is not sufficient.

**Support your users:** If the version of GIMP that you are selling has been modified by you in any way, you should inform your users and try to handle the support requests related to that version. Providing good support is another way to make your customers happy.

**Give back to the community:** If the software created by many volunteers helps your business, it would be nice to return the favour by helping the developers. You can contribute by sending some improvements to the code or by sponsoring some events such as the GIMP developer’s conference. This is not required but happy developers are more likely to create a better product that you can sell later.

While open source software can be obtained free, there are also some

issues involved when using them. One is the frequency of updates, which depends solely on the developers. Frequent updates are preferred so that the software remains useful. Another issue is the stability of the software. Business critical software must be stable and bug-free. Compatibility with proprietary software used by business partners is another issue. A company must be able to open a document sent by a business partner who uses proprietary software. **END** 

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The author is the chairman and MD of MultiSpectra Consultants, Kolkata. The company has developed what it calls the MultiSpectra OS, which basically consists of Ubuntu Linux with LibreOffice, Scribus, GIMP, Inkscape, Firefox, Thunderbird and VLC media player.